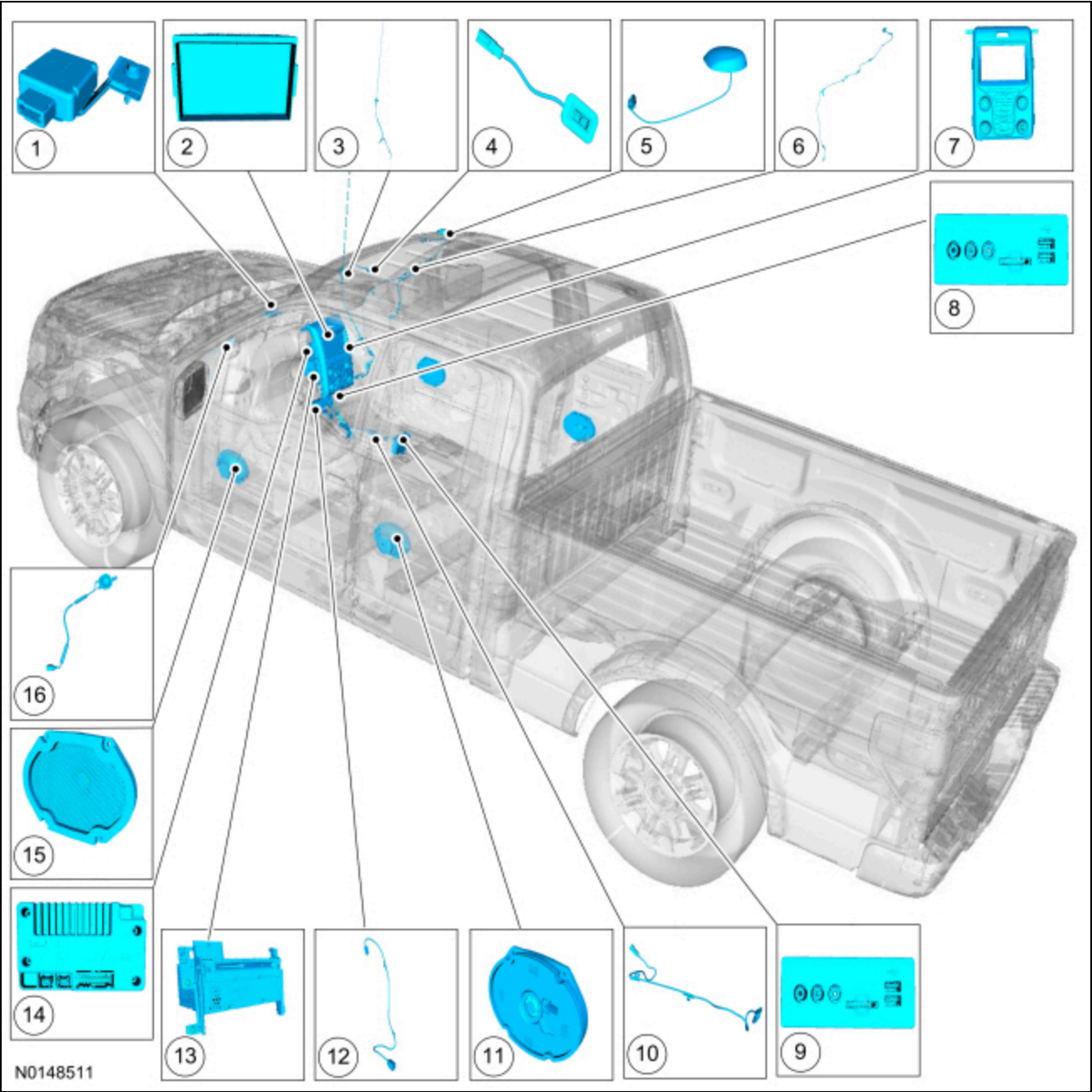


Information and Entertainment System

 [Ford Ecat Electronic Parts Catalog](#)

Component Location



Item	Description	Comments
1	GPSM	Located in the top center area of the instrument panel
2	FDIM	Attached to the APIM
3	AM/FM/HD Radio® antenna and AM/FM/HD Radio® coaxial cable	HD Radio® (if equipped)

Item	Description	Comments
4	SYNC® microphone	Located in the headliner
5	Satellite radio antenna	If equipped
6	Satellite radio antenna coaxial cable	If equipped
7	FCIM	-
8	Media hub	Located in the instrument panel (with split bench)
9	Media hub	Located in the console (with front captain chairs)
10	USB cable	Located in the console (with front captain chairs)
11	Rear door speaker	-
12	USB cable	Located in the instrument panel (with split bench)
13	ACM	-
14	APIM	Attached to the FDIM
15	Front door speaker	-
16	A-pillar speaker	LF shown, RF similar

Overview

The audio system consists of the following:

Main Components

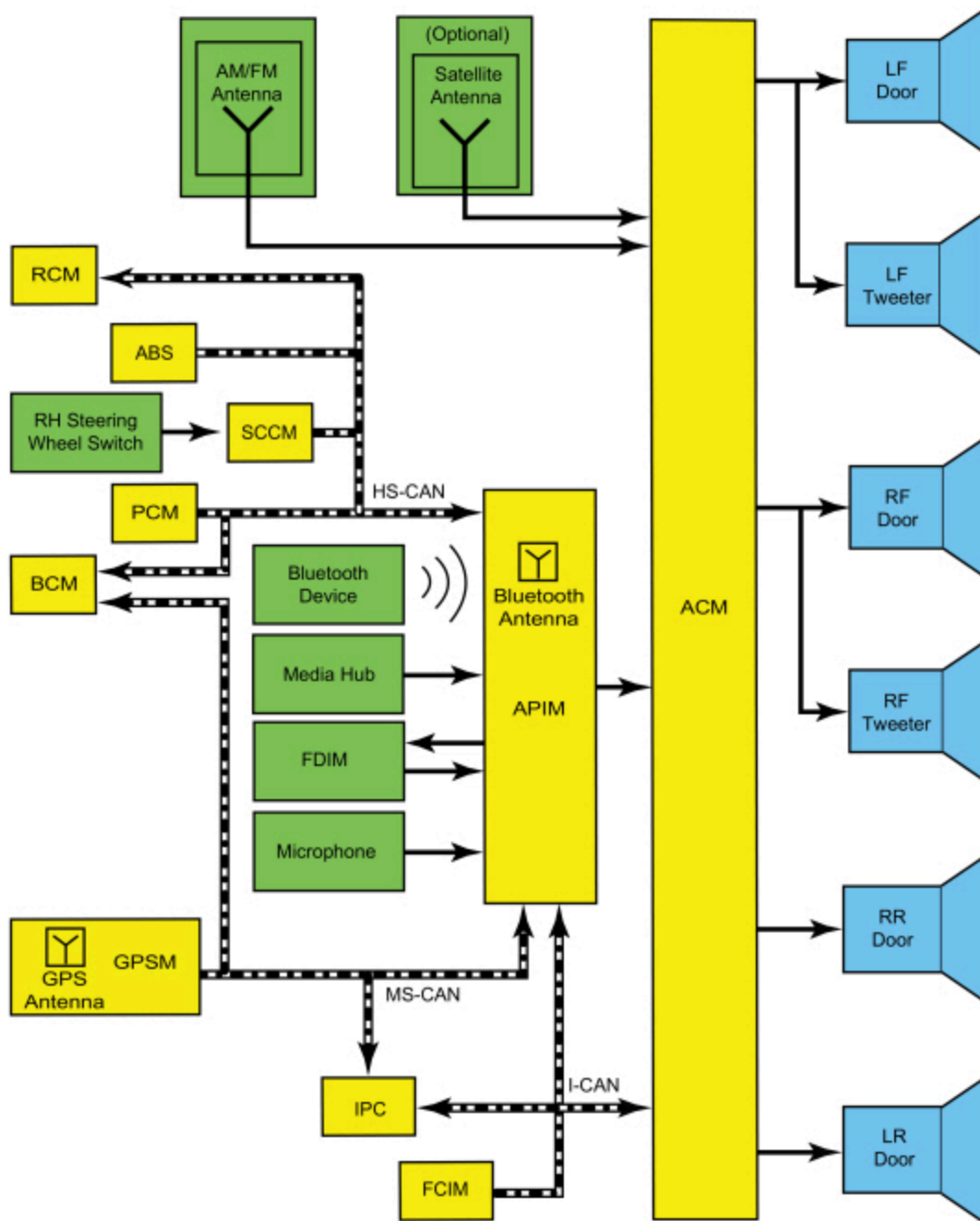
- 6 speakers
- 8-inch (203 mm) FDIM touchscreen
- AM/FM single CD ACM
- APIM
- FCIM
- GPSM
- Media hub
- Redundant steering wheel controls
- SYNC® microphone

Main Features

- Optional HD Radio®
- Optional navigation
- Optional satellite radio
- SYNC®

System Operation

System Diagram



N0148607

Network Message Chart

NOTE: A missing network message is indicated by a U-code DTC . When a missing message DTC (U-code) sets, it is important to look for symptoms present in the audio system and throughout the vehicle, and to review the complete message list to determine which other modules rely on the same message. Refer to CAN Module Communication Message Chart in Section 418-00. When the other modules have been identified, run the self-test for those modules. If the same message is missing from those modules, the identical, or similar lost communication DTCs may be set in those modules. Confirmation of missing messages common to multiple modules can indicate that the originating module is the source of the concern, or that the communication network may be at fault.

It is very important to understand the following:

- *Where the input originates*
- *All of the information necessary for a feature to operate*
- *Which module(s) receive(s) the input or command message*
- *Which module controls the output of the feature*
- *If the module that receives the input controls the output of the feature, or whether it outputs a message over the CAN to another module.*

Module Network Input Messages - ACM

Broadcast Message	Originating Module	Message Purpose
Belt-Minder® audio mute	IPC	Used to mute speaker output so the Belt-Minder® tone can be more easily heard. This message only applies to vehicles with a MyKey® enabled and in use. The audio system is muted until the safety belt(s) are buckled.
FCIM set volume	FCIM	Used to set the ACM volume from the FCIM .
Media player display request	APIM	Used to display detailed folder and track information on the FDIS from the ACM for the different audio system operating modes when the appropriate FCIM button is pressed.
MyKey® volume limit	IPC	Used to limit the maximum audio system volume when a MyKey® restricted key is in use.
Navigation audio settings	APIM	Used to set the navigation command volume.
SYNC® alerts	APIM	Used for text message, news, sports, weather, satellite radio artist/title favorites (if activated), and traffic alert notifications.

Module Network Input Messages - APIM

Broadcast Message	Originating Module	Message Purpose
ACM radio functions	ACM	Used to display the ACM radio functions such as seek, tune and AM/FM stations on the FDIS .
Airbag status	RCM	Used to monitor airbag deployment status for 911 assist.
Audio source	ACM	Used to display the selected audio source on the FDIS .
CD load/eject	ACM	Used to display the loaded or ejected CD status on the FDIS .
Date and time	IPC	Used to display the current date and time on the FDIS .
Door ajar status	RCM	Used to determine the door ajar status for the accessory delay feature.
FCIM button press	FCIM	Used to indicate when a button is pressed on the FCIM so the audio system can make the desired setting change.
Front seat belt latched status	RCM	Used to mute speaker output so the Belt-Minder® tone can be more easily heard. This message only applies to vehicles with a MyKey® enabled and in use. The

Broadcast Message	Originating Module	Message Purpose
		audio system is muted until the safety belt(s) are buckled.
Ignition key type	BCM	Used to determine if a MyKey® is enabled and in use.
Ignition status	BCM	Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for APIM operating modes and fault reporting.
Illumination dimming level	BCM	Used to control the backlight intensity of the FDIM based on the dimmer switch input.
Language selection	IPC	Used to display information on the FDIM in the selected language.
Navigation rolling wheel count	ABS module	Used to provide more accurate vehicle position tracking when the GPS signal is temporarily unavailable.
Seat belt buckle status	RCM	Used to mute speaker output so the Belt-Minder® tone can be more easily heard. This message only applies to vehicles with a MyKey® enabled and in use. The audio system is muted until the safety belt(s) are buckled.
Steering wheel switch media	SCCM	Used to set the audio system listening mode (AM/FM, CD, etc.) from the steering wheel switch.
Steering wheel switch seek left	SCCM	Used to access the next strong station to the left, to restart an audio track, to access the previous satellite radio station preset (if equipped), or the previous audio track from the steering wheel switch.
Steering wheel switch seek right	SCCM	Used to access the next strong station to the right, to access the next satellite radio station preset (if equipped), or to access the next audio track from the steering wheel switch.
Steering wheel switch voice	SCCM	Used to access or exit the voice command feature from the steering wheel switch.
Steering wheel switch volume down	SCCM	Used to decrease the audio system volume from the steering wheel switch.
Steering wheel switch volume up	SCCM	Used to increase the audio system volume from the steering wheel switch.

Module Network Input Messages - FCM

Broadcast Message	Originating Module	Message Purpose
Door ajar status	BCM	Used to determine the door ajar status for the accessory delay feature.
Ignition status	BCM	Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for FCM operating modes and fault reporting.
Illumination dimming level	IPC	Used to control the backlight intensity of the FCM based on the dimmer switch input.

Module Network Input Messages - GPSM

Broadcast Message	Originating Module	Message Purpose
Ignition switch position	BCM	Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for GPSM operating modes and fault reporting.
Selector lever (PRNDL) status	BCM	Used to verify the gear lever position for vehicle direction data.

Accessory Delay Feature

The accessory delay feature allows the audio system to be operated for a preset period of time after the ignition is turned off and a front door has not been opened.

AM/FM Radio

The AM/FM fender-mounted antenna receives radio waves and sends them through the AM/FM antenna coaxial cable to the ACM . The radio waves are converted into fluctuating AC voltage and sent as left and right analog audio signals to the speakers.

APIM Programming

The APIM requires specific programming procedures for correct operation. APIM programming is required when:

- The APIM is replaced
- Directed by a TSB. Updates to resolve SYNC® performance concerns can only be performed when directed by a TSB.
- Feature upgrades are requested. Occasional feature upgrades and/or new applications can be programmed when they become available. This type of programming is available upon customer request and expense.

There are 3 types of APIM programming available:

- Module Replacement — Used when the APIM has been replaced. This type of programming performs the required provisioning (restores software for newly installed hardware).
- Standard — Used to update the APIM and any currently installed Applications to the latest software level.
- Custom — Used to show all the available software options (including applications that may not have been installed previously) that can be programmed into the APIM . Use this selection when upgrades become available or as directed by a TSB.

There are 3 types of software installation methods:

- Full Flash (Force CIP) — A CIP program is the entire software package for the APIM . Performing a Full Flash erases the existing APIM software and installs the complete program. A Full Flash should only be performed if a Service Pack is not available. If a Full Flash is performed, all applications must be re-installed. Performing a Full Flash by selecting Force CIP requires additional programming time and is not necessary unless directed by a TSB.
- Service Pack — A Service Pack update only provides the necessary software to update the existing APIM program to a new level. For example, a Service Pack changes the software program from an 'AA' to an 'AB' version level. It does not delete or overwrite the existing APIM software. A Service Pack is a much smaller program than a Full Flash and requires less downloading and programming time. A Service Pack software installation should always be performed instead of a Full Flash, if possible.
- Applications — Applications are special software programs that may be installed on the APIM . Applications necessary for each vehicle are determined by the vehicle build information stored in the Ford online database. Selecting individual applications for installation may not be necessary every time the APIM is programmed.

NOTE: VIP programming is not selectable because the VIP is configured automatically during programming.

There are some general programming guidelines that are applicable to multiple types of programming:

- Full Flash/Force **CIP**.
 - The media hub must be removed for programming.
 - The diagnostic scan tool and SYNC® Re-Flash Kit (or equivalent), are required to complete the flash procedure.
 - All previously paired phones are deleted from the SYNC® system.
 - The file needs to be downloaded from the Ford online database. Allow for extra time in order to complete the download. The compressed files are between 200 and 400 megabytes (MB), and once downloaded, are extracted to the full size of approximately 2 gigabytes (GB).
 - Once diagnostic scan tool programming is complete, applications need to be installed with the **USB** flash drive. This should be accomplished by reinstalling the media hub and connecting the **USB** flash drive to the media hub.
- Service Pack and Applications
 - A **USB** flash drive with a minimum 1 gigabyte (GB) storage capacity is required.
 - The file is downloaded from the Ford online database onto the flash drive. The files are then transferred via the flash drive to the **APIM** through the vehicle **USB** port.

To perform module replacement programming, refer to Accessory Protocol Interface Module (APIM) Module Replacement Programming

To perform standard programming, refer to Accessory Protocol Interface Module (APIM) Standard Programming

To perform custom programming, refer to Accessory Protocol Interface Module (APIM) Custom Programming

Audible Prompts

The **APIM** receives stereo inputs from connected mobile phones and mono inputs from the SYNC® microphone.

Stereo and mono inputs, voice prompts, the **TTS** feature, ringtones, and any audio received through a connected mobile phone are transmitted to the **ACM** as stereo or mono sound.

The **TTS** feature speaks information so that it does not have to be read from the display.

Audio Extended Play

This feature allows the audio system to operate with the ignition OFF, regardless of door latch position. When the power button on the **FCIM** is pressed, audio system functionality is enabled and remains active for a period of 20-60 minutes, depending on configuration settings. To turn the audio system off, the power button on the **FCIM** is pressed. Audio extended play will not function if the battery load shed is active.

Battery Load Shed

The **BCM** uses the battery current sensor to monitor the battery state of charge. The battery current sensor is attached to the battery ground cable. With the vehicle OFF and the ignition in ACC or ON, a load shed message is sent over the **CAN** when the **BCM** determines that the battery state of charge is below 40%, 45 minutes have elapsed, or 10% of the charge has been drained. This message turns off the audio system to save the remaining battery charge. Under this condition, SYS OFF TO SAVE BATT is displayed on the centerstack infotainment display to notify the driver that battery protection actions are active. To clear the load shed state, start the vehicle.

Bluetooth Mode

Bluetooth is a secure, short-range radio frequency that allows devices to communicate wirelessly through radio waves. The operating range of a Bluetooth signal is a maximum of 9.75 m (32 feet).

The Bluetooth interface can accommodate both Bluetooth-enabled mobile phones and Bluetooth-enabled media devices. The SYNC® system allows interaction with several types of customer Bluetooth devices, including mobile phones and media devices. The **APIM** contains an on-board Bluetooth chipset, which enables certain wireless devices to interact with the system. Any Bluetooth device used with the SYNC® system must first be paired with the system before it is operational.

Only one Bluetooth phone and one Bluetooth media device can be connected to the system at any one time. If an additional device of either type is paired with the system and made active, the **APIM** disconnects any active connection and establishes

a connection with the new device.

When a new Bluetooth device is added, the **APIM** and the Bluetooth device must be paired together. Most Bluetooth devices can pair with the SYNC® system, although functionality may vary. To determine if a Bluetooth device is supported, verify the customer device is on the compatibility list for the current **APIM** software level.

Pairing a Bluetooth device is accomplished through the "Add Device" selection of the phone menu. When pairing a device, the SYNC® system generates a unique **PIN** that must be entered on the Bluetooth device in order for the pairing process to be successful. There are also some device-specific actions that must take place. For information on the pairing process, refer to the Owner's Literature.

It is important to understand that not all mobile phones have the same level of features when interacting with the SYNC® system. For a list of compatible phones, refer to [SyncMyRide website](#).

Clock Operation

The **APIM** broadcasts a date/time status through CAN traffic messages for other modules to use whenever the CAN bus is awake and transmitting information. A default date/time value may be used by modules on the network if the date/time information is not available from the **APIM**.

Composite Audio-Video Input Mode

The composite audio and video inputs consist of the RCA jack connections that are used for playing video or audio content from an external device (such as a gaming system). The media hub contains an internal composite audio/video translator connected to the external RCA jacks, which outputs video and audio signals to the **APIM** via a shielded audio and video harness.

HD Radio® (If Equipped)

HD Radio® is a free to the public digital radio broadcast. **HD Radio®** channels duplicate certain analog FM channels, and some subdivide (multicast) the digital signal into multiple channels that do not have analog equivalents.

The AM/FM fender-mounted antenna receives radio waves containing data (station call letters, artist names, song titles, etc.) and digital **HD Radio®** audio. The radio waves are sent through the AM/FM antenna coaxial cable to the **ACM**, where the data is separated from the audio. The data is sent to be displayed on the centerstack infotainment display, and the audio is converted into fluctuating AC voltage and sent as left and right analog audio signals to the speakers.

MyKey® Audio Operation

When a MyKey® is in use, the following restrictions are activated:

- The audio system is muted until the safety belts are buckled when the Belt-Minder® feature is activated. The MyKey® "BUCKLE UP TO UNMUTE AUDIO" is displayed in the information display.
 - This is a standard setting and cannot be changed.
- Satellite radio adult content stations are blocked by being muted.
- If "Always On" is selected in the MyKey® menu, 911 assist and the Do Not Disturb features are enabled and cannot be disabled.
- The audio system volume is limited to 45%. In an attempt to exceed the limited volume, the MyKey® VOLUME LIMITED message is displayed in the centerstack infotainment display.
 - This is an optional setting.

Refer to the Owner's Literature for more information on MyKey®.

Navigation

The navigation system guides the user to a pre-entered destination. Map data is read from the map data SD card plugged into the media hub. The **APIM** calculates route information based on **GPS** data received by the **GPSM**. The **APIM** also uses vehicle speed and transmission gear selected signals received through the network to detect vehicle speed and direction, resulting in more accurate navigation tracking. The navigation display is shown on the **FDIM**. Installing different size tires other than what is indicated on the vehicle placard may cause the navigation system to be inaccurate.

The voice recognition system allows the user to interface with the system without using the touchscreen. A microphone located in the headliner provides a direct input to the **APIM** .

Satellite Radio (If Equipped)

Satellite radio is a subscription-based service.

The satellite radio roof-mounted antenna receives signals containing data (station call letters, artist names, song titles, etc.) and digital satellite radio audio. These signals are sent through the satellite radio antenna coaxial cable to the satellite radio receiver built into the **ACM** . The **ACM** separates the data from the audio. The data is sent to be displayed on the centerstack infotainment display, and the audio is converted into fluctuating AC voltage and sent as left and right analog audio signals to the speakers.

SD Card Mode

The SD card slot can be used for playing audio content that is stored on a SD card.

Image files can be saved to the **APIM** from a SD card in order to be used as wallpaper backgrounds displayed on the **FDIM** .

SD card data is transmitted to the **APIM** through the **USB** cable.

SIRIUS Data Services - If Equipped

SIRIUS Data Services is a subscription-based service and requires a separate subscription from the satellite radio.

The combination **HD** /satellite radio roof-mounted antenna receives digital signals and sends them to the satellite radio receiver through the satellite radio antenna coaxial cable. The satellite radio receiver is built into the **ACM** . The signals are then sent to the **APIM** , and then to the **FDIM** touchscreen to be displayed.

SIRIUS Data Services allows the customer to receive updates for:

- Fuel Prices
- Movies (movie listings, start times, ratings)
- Sports (in game and final scores, weekly schedules)
- Stocks
- Ski Conditions
- Traffic (traffic speed, accidents, construction, road closings)
- Weather (current weather, forecasts, radar images)

Some services may be available separately while others are only available as packages.

Speed Compensated Volume

The **ACM** adjusts the audio system volume based on the **VSS** signal to compensate for speed and wind noise.

When a MyKey® is in use and the MyKey® radio volume limiter is enabled, the speed compensated volume is disabled.

Steering Wheel Switch Functions

The **RH** 8-way steering wheel switches operate the volume +/-, seek +/-, and the audio system **SYNC**® functions (media, voice, and phone).

SYNC® Services

SYNC® Services is a subscription-based service. For **SYNC**® Services subscription information,, refer to [SyncMyRide](#).

The customer's mobile phone is used to access and download the **SYNC**® services. **SYNC**® services information received by the customer's mobile device is sent to the **APIM** .

The **GPS** antenna, integral to the **GPSM** , is used to detect the location and direction of the vehicle.

SYNC® services includes the following features:

- Business search
- Personalized text alerts
- Real-time traffic updates
- Turn-by-turn directions

SYNC® services allows the customer to receive updates for:

- Entertainment
- News
- Sports
- Stocks
- Travel
- Weather

SYNC® System

The SYNC® system is a hands-free communication and entertainment system that supports many features. All features are not available with every phone.

The SYNC® system provides the ability to:

- Connect media devices (such as an iPod® or USB device) in order to play audio files.
- Control the entertainment system, climate control system, ambient lighting system, and several other vehicle systems using the touchscreen or voice controls.
- Initiate an emergency call (911 Assist™) when the airbags deploy.
- Play media files via a paired Bluetooth-enabled audio device.
- Produce vehicle health reports.
- Send and receive phone calls via a paired Bluetooth-enabled phone.
- Send and receive text messages via a paired Bluetooth-enabled phone.
- Wi-Fi signal receiver for in-car Internet access (requires customer-supplied USB modem).

USB Mode

The USB port can be used for connecting a media device (such as an iPod®) with the device's available cable, or for directly plugging in a portable mass storage device, such as a thumb drive. When playing media files stored on a mass storage device, the SYNC® system only plays files that do not have DRM protection. The USB port can also be used for uploading vehicle application upgrades. The USB port is powered by the APIM, so no external power source is needed to power a device plugged into the USB port if the device supports this feature.

In addition to audio information, metadata (information such as artist, album title, song title, and genre) may also be sent to the APIM from a device plugged into the USB port. The APIM uses the metadata to create indexes that can be used to sort for particular music, based on customer preference. Not all USB devices can send metadata to the APIM. When a new media device is connected to the SYNC® system, the APIM automatically indexes the information. This may take several minutes (depending on the amount of data on the device), and is considered normal operation. When a device that was previously connected to the SYNC® system is reconnected, the APIM updates the index (rather than creating a new one), which reduces the amount of time needed to index the device.

Image files can be saved to the APIM from a USB device. These images can be displayed on the FDIS as wallpaper backgrounds.

Voice Recognition

Voice recognition is used for many vehicle functions, including audio system and climate controls. The microphone relays the microphone input to the APIM through dedicated wiring. The TTS and voice prompt features speak certain text information and interaction requests to minimize driver distraction by having to look at the audio system display while driving. The ringtone alerts the driver to an incoming call. The microphone is also used to detect outgoing audio during a phone call.

Audible prompts can range from a simple tone to more elaborate spoken text, based on the customer setting. When interaction mode is set to standard, detailed guidance is provided. When interaction mode is set to advanced, most prompts are tones only and minimal audible guidance is provided. Refer to the Owner's Literature for further information on voice interaction.

The audio signals for the **TTS** and voice prompt features, ringtones, and audio from the outside device during a phone call, are sent from the **APIM** to the **ACM**.

Component Description

ACM

The **ACM** can be operated with the ignition in RUN, ACC, or OFF.

The **ACM** receives radio waves containing audio from the antenna coaxial cable(s) and converts them into fluctuating AC voltage. The **ACM** receives audio signals from the **APIM** in the form of fluctuating AC voltage. The **ACM** processes these inputs and sends left and right analog audio signals to the speakers as fluctuating AC voltage.

If equipped with satellite radio or **HD** Radio®, the **ACM** will also receive radio waves containing data through the antenna coaxial cable(s). The data is sent to be displayed on the centerstack infotainment display.

The **ACM** requires **PMI** when it is replaced.

Antenna

The AM/FM fender-mounted antenna receives radio waves containing audio and sends them through the AM/FM antenna coaxial cable to the **ACM**. The antenna mast is used to amplify the AM/FM radio waves to improve reception.

If equipped with **HD** Radio®, the AM/FM fender-mounted antenna receives radio waves containing data and digital audio and sends them through the AM/FM antenna coaxial cable to the **ACM**.

If equipped with satellite radio, the satellite radio roof-mounted antenna receives signals containing data and digital audio. These signals are sent through the satellite radio antenna coaxial cable to the satellite radio receiver built into the **ACM**.

APIM

Audio signals received by the **APIM** are sent to the **ACM** as fluctuating AC voltage.

The **APIM** consists of 2 internal modules: the **CIP** and the **VIP**. The modules are not replaceable individually, but can be flashed independently, if required.

The **CIP** interfaces with all of the inputs to the **APIM**. It contains an analog-to-digital-to analog converter, as well as the Bluetooth chipset. Any Application upgrades that are available to the consumer are loaded directly to the **CIP** through the **USB** port.

The **VIP** provides an interface between the **CIP** and the vehicle. Its main functions are controlling the **APIM** power management and translating inbound and outbound **CAN** signals. In addition, the **VIP** queries the modules on the network to retrieve any **DTCs** when a vehicle health report is requested.

FCIM

The **FCIM** can be operated with the ignition in RUN, ACC, or OFF.

The **FCIM** audio system buttons provide one of the methods in which the customer interacts with the infotainment system.

The **FCIM** is separate from the **ACM**.

EDIM

The **EDIM** can be operated with the ignition in RUN, ACC, or OFF.

The **FDIM** touchscreen provides one of the methods in which the customer interacts with the infotainment system.

The **FDIM** plugs directly into the **APIM** , and therefore does not communicate directly on any network, and no external circuits are connected.

Global Positioning System Module (GPSM)

The Global Positioning System Module (GPSM) provides vehicle location for real-time traffic reports and re-routing, and for identifying vehicle location in the event of a collision.

For vehicles with navigation, the Global Positioning System Module (GPSM) acts as the antenna for the navigation system.

Vehicle location information is broadcast from the Global Positioning System Module (GPSM) to the **APIM** .

Media Hub

The media hub allows for various audio and video devices to play content through the vehicle speakers and to be viewed in the **FDIM** .

The media hub receives inputs from:

- 2 **USB** ports
- 1 Secure Digital (SD) card slot
- 1 set of component RCA jacks

Steering Wheel Switches

The **RH** steering wheel switches contain series of resistors. Each steering wheel switch function corresponds with a specific resistance value within the switch.

The **SCCM** sends a 5-volt reference voltage to the **RH** steering wheel switch on the input circuits. When a switch is pressed, the voltage is routed through the corresponding resistor(s), and then to ground through the **SCCM** . The **SCCM** monitors the resultant voltage drop to determine which switch was pressed. The voltage drop varies depending upon the resistance of the specific switch pressed.

The **SCCM** sends a network message to the **APIM** for the SYNC® switches and to the **ACM** for the redundant audio controls.

SYNC® Microphone

The SYNC® microphone receives SYNC® system voice commands and outgoing audio during a phone call. These signals are sent to the **APIM** .